

Client Setup

Overview

The client side of the DarshanEase application is developed using **React.js**, **TypeScript**, and **Tailwind CSS**. It provides a responsive and user-friendly interface for devotees and administrators to interact with the system.

Prerequisites

Before setting up the client application, ensure the following software is installed:

- Node.js (v18 or later)
- npm (Node Package Manager)
- Visual Studio Code
- Git (optional)

Step 1: Create the React Project

Open a terminal and run the following command:

```
npm create vite@latest frontend -- --template react-ts
```

Move into the project folder:

```
cd frontend
```

Step 2: Install Dependencies

Install all required packages:

```
npm install
```

Step 3: Install Tailwind CSS

Install Tailwind CSS and its dependencies:

```
npm install tailwindcss @tailwindcss/vite
```

Configure the Vite plugin in **vite.config.ts**:

```
import { defineConfig } from 'vite'
```

```
import react from '@vitejs/plugin-react'
import tailwindcss from '@tailwindcss/vite'
```

```
export default defineConfig({
  plugins: [
    react(),
    tailwindcss(),
  ],
})
```

Import Tailwind CSS in **src/index.css**:

```
@import "tailwindcss";
```

Step 4: Install Additional Packages

Install commonly used libraries for routing, HTTP requests, icons, and notifications:

```
npm install react-router-dom axios react-icons react-hot-toast
```

(Optional) Install development tools:

```
npm install -D eslint prettier
```

Step 5: Create the Folder Structure

Organize the client project as follows:

```
frontend/
├── public/
├── src/
│   ├── assets/
│   ├── components/
│   ├── pages/
│   ├── layouts/
│   ├── routes/
│   ├── services/
│   ├── hooks/
│   ├── context/
│   ├── utils/
│   ├── types/
│   ├── App.tsx
│   ├── main.tsx
│   └── index.css
└── package.json
```

└─ vite.config.ts
└─ tsconfig.json

Step 6: Configure Environment Variables

Create a `.env` file in the frontend root directory.

`VITE_API_URL=http://localhost:5000/api`

The application can access this value using:

```
const API_URL = import.meta.env.VITE_API_URL;
```

Step 7: Start the Client Application

Run the development server:

```
npm run dev
```

Example output:

Local: `http://localhost:5173/`

Open the displayed URL in your web browser to access the DarshanEase client application.

Client Responsibilities

The client application is responsible for:

- User registration and login
- Displaying temple information
- Showing available darshan slots
- Booking darshan tickets
- Viewing booking history
- Managing user profiles
- Admin dashboard access
- Sending API requests to the backend
- Displaying responses received from the server

Technologies Used

Technology	Purpose
React.js	User Interface Development
TypeScript	Type-safe JavaScript
Vite	Build Tool and Development Server
Tailwind CSS	Responsive UI Styling
React Router	Client-side Routing
Axios	API Communication
React Icons	Icons
React Hot Toast	Notifications

Conclusion

The client setup establishes the frontend environment for DarshanEase using modern web development tools. With React, TypeScript, Tailwind CSS, and Vite, the application delivers a fast, responsive, and maintainable user interface that communicates efficiently with the backend through REST APIs.